# PAVITHRA S

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## CAREER OBJECTIVE

To build a meaningful career in Embedded Systems by turning ideas into real-world solutions through programming and hardware integration. I look forward to working on real-world projects, contributing innovative ideas and designs, and continuously improving through practical experience while delivering reliable and effective solutions that add value to the organization.

## WORK EXPERIENCE

- Currently undergoing hands-on technical training program – **Advanced Embedded Systems Course** at Emertxe Information Technologies (http://www.emertxe.com), Bangalore
- This course is Government of India certified program, aligned with **Skill India** / NSDC under Electronics Sector Skill Council of India (http://www.essc-india.org) - **Embedded Software Engineer QP ELE /Q1501**

## TECHNICAL SKILLS

- Programming Languages:
  - Shell scripting
  - Advanced C programming
  - OOP using C++
  - Data structures
- Embedded controllers:
  - Hands-on working with GPIOs, Analog I/Os, Memory usage, interfacing, character LCD
  - Peripherals usage - Timers, Counters and Interrupts
  - Communication protocols - UART, SPI, I2C etc
- Embedded platforms:
  - Distributions - Linux (Ubuntu)
  - PIC16F877A board
- Development environment and tools:
  - Dev environment: Vim, Makefiles, MPLAB, PicSimLab
  - Compilers: GCC, XC8, ARM-Linux-gcc
  - Debuggers: GDB

## COURSE WORK

- Microprocessor
- Digital Electronic
- VLSI

- Digital signal processing
- C programming
- C++
- Data structures

## EDUCATION

- B.E (ECE), Mepco Schlenk Engineering College, CGPA – 6.61, 2021-2025
- Class – XII, State Board, 85.68%, 2021
- Class – X, State Board, 83.4%, 2019

## PROJECTS AT EMERTXE

**Project Number:1**

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|---|---|
| **Title** | Address Book |
| **Project brief** | The objective of this project was to develop an Address Book application to store and manage contact details. The program allows users to add, search, edit, delete, and list contacts such as name, phone number, and email address. The data is stored in a file so that the contacts remain saved even after the program is closed. The application uses structured programming concepts to organize and manage contact records efficiently. |
| **Technologies used** | Advanced C – Structures, File operations, Pointers, Functions, Arrays, Command line interface |
| **Key challenges & Learnings** | • Understanding how to store and manage multiple contact records using structures<br>• Implementing file handling to save and retrieve contacts from storage<br>• Handling search, update, and deleting operations efficiently<br>• Faced challenges while managing file read/write operations and struct data handling which was resolved through debugging and practice |

**Project number:2**

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| **Title** | Steganography |
| **Project brief** | The objective of this project was to hide a secret text message inside an image file using the Least Significant Bit (LSB) technique. The program encodes the length of the secret message and then embeds the text data into the LSB of the image bytes of a BMP file without affecting the visible quality of the image. The decoding process extracts the length first and then retrieves the hidden message bit by bit from the encoded image to obtain the original secret text. |
| **Technologies used** | Advanced C – File operations, Pointers, Bitwise operations, Functions, Makefiles, Command line arguments |
| **Key challenges & Learnings** | • Understanding the structure of BMP image files including header and pixel data<br>• Implementing LSB encoding technique to hide data without changing the appearance of the image<br>• Performing bitwise manipulation for embedding and extracting secret information<br>• Faced challenges while handling file operations and bitwise data processing which were resolved through debugging and testing |

**Project number:3**

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| **Title** | Arbitrary Precision Calculator |
| **Project brief** | Developed a calculator capable of performing arithmetic operations on very large numbers that exceed standard C data type limits. The project uses **Double Linked Lists** to store digits and implements operations like addition, subtraction, multiplication, and division. |
| **Technologies used** | Advanced C, Data Structures - Double Linked List, Dynamic Memory Allocation, Pointers |
| **Key challenges & Learnings** | • Handling arithmetic operations for numbers with hundreds of digits, managing carry/borrow across multiple nodes, and implementing efficient division using linked list representation.<br>• Also found difficulties while performing division operations.<br>• Gained strong understanding of Double Linked Lists, dynamic memory management, and implementing large number arithmetic algorithms using data structures. |

**Project number:4**

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| **Title** | Inverted Search |
| **Project brief** | Developed a search system that builds an inverted index from multiple text files to enable fast searching of words across documents. The program reads input files, parses words, and stores them using hashing and linked lists. Each word is mapped to the files where it appears along with the occurrence count. The system supports database creation, searching, displaying, saving, and updating the index to improve search efficiency. |
| **Technologies used** | Advanced C, Data Structures - Hashing, Linked List, File Handling, Dynamic Memory Allocation, String Processing |
| **Key challenges & Learnings** | • Designing an efficient hash table indexing mechanism for quick word lookup.<br>• Managing dynamic linked lists to store words and file details.<br>• Handling duplicate words across multiple files and maintaining accurate word counts.<br>• Implementing file validation and database update functionality.<br>• Gained practical experience in implementing hash tables and linked lists for efficient data storage and retrieval.<br>• Improved understanding of file handling, dynamic memory allocation, and search optimization techniques in C. |

## ACADEMIC PROJECTS

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| **Title** | **Design of Approximate Adder Architectures for Image Blending using Verilog** |
| **Project brief** | Designed and implemented approximate full adder architectures using Verilog to improve power and area efficiency in digital circuits. The project applies approximate computing techniques to perform image blending, where two images are combined while maintaining acceptable image quality using error-tolerant arithmetic operations. |
| **Technologies used** | Modelsim, Matlab, Xilinx Vivado - Verilog HDL, MATLAB, Digital Electronics, VLSI Concepts Image Processing Techniques. |
| **Key challenges & Learnings** | • Designing approximate adder logic while maintaining acceptable accuracy. |

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| | • Implementing image blending and evaluating image quality using performance metrics.<br>• Developed understanding of approximate computing concepts for low-power digital system design.<br>• Learned how arithmetic circuit design impacts power, delay, and hardware area in VLSI systems.<br>• Enhanced ability to analyze circuit performance using error and image quality metrics. |

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| **Title** | **Automatic Railway Gate Control System** |
| **Project brief** | Designed and implemented an automated railway gate control system that detects an approaching train and automatically opens and closes the railway gate to improve safety and reduce manual intervention. The system uses sensors and a microcontroller to control the gate mechanism. |
| **Technologies used** | Arduino, Arduino IDE, Sensors, Servo Motor, Basic Electronics |
| **Key challenges & Learnings** | • Initial implementation using 8051 microcontroller did not function as expected due to hardware interfacing and timing issues.<br>• Troubleshooting sensor detection and motor control logic.<br>• Gained practical experience in microcontroller interfacing and embedded programming.<br>• Learned debugging hardware–software integration issues and adapting the design using Arduino for successful implementation. |